

Attention Is All You Need - Intuition and Explanation

1 Why Attention Was Needed (The Problem First)

Before Transformers, we had:

- RNN
- LSTM
- GRU

Problem:

If sentence is long:

“The cat that was adopted by the family who lived near the river was very playful.”

To understand “was very playful”, the model must remember “The cat”.

RNNs pass information step by step → long path → vanishing gradient → weak long-term memory.

Attention solves this by allowing direct connection between any two words.

Instead of remembering everything sequentially, the model can look directly at relevant words.

2 Attention Mechanism (Core Idea)

Imagine you are translating:

“The animal didn’t cross the street because it was too tired.”

When translating “it”, you must decide:

- Does “it” refer to animal?
- Or street?

Attention says:

When predicting a word, look at **ALL input words** and decide which are important.

Mathematically:

$$\text{Score} = \text{similarity}(\text{query}, \text{key})$$
$$\text{Attention} = \text{softmax}(\text{score})$$
$$\text{Output} = \text{weighted sum of values}$$

Simplified explanation:

Each word creates:

- **Query** (What am I looking for?)
- **Key** (What do I contain?)
- **Value** (Actual information)

If query matches key $\rightarrow$ strong attention.

Simple Example

Sentence: “I love deep learning”

When computing representation for “love”:

- How related am I to “I”?
- How related am I to “deep”?
- How related am I to “learning”?

Maybe “deep” and “learning” get higher weights.

So new representation of “love” becomes weighted mix of all words.

That’s attention.

3 Self-Attention (Very Important)

Self-attention means:

Words attend to other words in the SAME sentence.

Each word interacts with every other word.

Example:

“The bank of the river”

Word “bank” could mean:

- Financial bank
- River bank

Self-attention allows “bank” to look at “river” $\rightarrow$ meaning becomes clear.

So instead of static embedding, we get **contextual embedding**.

That’s why LLM embeddings are powerful.

4 How Attention is Computed (Intuition Version)

We create matrices:

$$Q = XW_q, \quad K = XW_k, \quad V = XW_v$$

Then:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d}}\right)V$$

Why divide by $\sqrt{d}$?

Because when dimension grows, dot products grow large $\rightarrow$ softmax becomes sharp $\rightarrow$ gradients unstable.

Scaling stabilizes training.

5 Multi-Head Attention

Instead of one attention mechanism, Transformer uses **multiple heads**.

Each head learns different relationships:

- One head learns grammar
- One head learns subject-verb relation
- One head learns positional relation

Outputs are concatenated.

Think of it like: multiple perspectives looking at same sentence.

6 Encoder vs Decoder

Transformer has two blocks:

Encoder

- Reads input
- Produces contextual embeddings
- Uses self-attention
- Used in: BERT, Classification, Embeddings

Decoder

- Generates output step by step
- Uses:
 - Masked self-attention (cannot see future words)
 - Cross-attention (attends to encoder output)
- Used in: GPT, Text generation

Key Difference:

Encoder = Understanding Decoder = Generating

- GPT = Decoder-only
- BERT = Encoder-only
- Original Transformer = Encoder + Decoder

7 Why Scaling Works (Very Important Concept)

LLMs follow **scaling laws**:

More:

- Parameters
- Data
- Compute

→ Better performance.

Why?

Because Transformers:

- Learn language patterns statistically
- Store massive knowledge in weights
- Approximate next-token probability better with more capacity

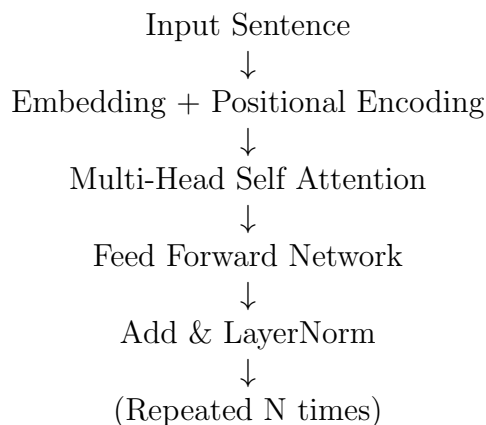
Emergent abilities appear when model becomes large enough.

Example:

- Small model → poor reasoning
- Large model → can solve multi-step math

It's not magic. It's pattern generalization at scale.

8 Full Transformer Architecture (Simple View)



That's encoder block.

Decoder adds:

- Masked attention
- Cross attention

9 Why Transformer Beat RNN

RNN	Transformer
Sequential	Parallel
Hard long memory	Direct connections
Slow training	Fast training
Vanishing gradients	Stable training

Parallelization was a game changer.

10 Big Intuition in One Line

Transformer = “A system where every word can directly talk to every other word and decide what matters.”

Final Mental Model

Think of it like a group discussion.

Each word:

- Asks questions (Query)
- Advertises itself (Key)
- Shares information (Value)

Important words influence others more.